

GreenCompass: Experiencing Urban Pocket Time as Playful Nature Connection through Context-Aware Gamification

Mingtao Wu

School of Computer Science and Engineering
Southeast University
Nanjing, China
220232308@seu.edu.cn

Ziteng Zhang

School of Computer Science and Engineering
Southeast University
Nanjing, China
zitengzhang@seu.edu.cn

Zhuying Li*

School of Computer Science and Engineering
Southeast University
Nanjing, China
zhuyingli@seu.edu.cn

Yan Wang

Exertion Games Lab
Monash University
Melbourne, Victoria, Australia
yan@exertiongameslab.org

Abstract

As urbanization accelerates, there is a growing lack of natural environments in cities, leading to a significant reduction in opportunities for people to interact with nature. In response, HCI researchers have increasingly explored methods to enhance human-nature interactions. However, much of this work is primarily focused on the improvement of the situational experience of engagement with nature. We believe that it is equally important to inspire and motivate people to reconnect with the natural world, fostering a deep relationship with their environment. To this end, we present GreenCompass, a mobile social game that encourages players to engage with nature during their pocket time by forming groups with friends to complete nature-related tasks. With GreenCompass, we aim to inspire further design opportunities that enhance human-nature interactions in urban environments.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction design*.

Keywords

Human-nature interaction; nature engagement; games; mobile social games

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*Corresponding author

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1 INTRODUCTION

Urbanization has increasingly distanced people from direct interaction with nature, with profound effects on mental health, physical well-being, and overall quality of life [13, 41]. This sense of disconnect is further exacerbated by urban environments that prioritize infrastructure over green spaces, making it difficult to reverse these trends [46]. In response, human-computer interaction (HCI) research has begun to focus more on human-nature interaction (HNI), highlighting its potential to restore the benefits of engaging with nature [2, 11, 26, 49]. Most of the work in human-nature interaction remains theoretical, exploring frameworks and discussions that conceptualize the relationship between humans and nature [2, 40, 48]. Although less common, practical applications often focus on improving natural experiences or users' understanding of the environment through technology [19, 23, 43]. However, interacting with nature transcends a single dimensional experience and involves multiple interrelated aspects that require a more holistic approach.

Masashi et al. [42] have proposed a unified conceptual framework for human-nature interaction that emphasizes the interconnectedness of experiencing nature, its drivers, consequences, and the feedback loops between these elements. Central to this framework are the drivers—opportunity, motivation, and capability—which are crucial for overcoming barriers to connecting with nature and fostering sustained engagement with the natural world. Inspired by this framework, we developed GreenCompass, a mobile social game that encourages players to engage with nature during their pocket time by forming groups with friends to complete nature-related tasks. GreenCompass provides context-aware task suggestions based on the user's time, location, and preferences, creating opportunities for users to be able to engage with nature easily. To increase motivation, GreenCompass incorporates gamification and social features, encouraging immediate and sustained engagement with nature. In addition, GreenCompass takes into account the user's capability to interact with nature by supporting different fitness levels and embedding educational elements to increase ecological knowledge. By integrating opportunity, motivation, and capability, GreenCompass aims to overcome urban barriers, stimulate nature engagement,

and foster lasting connections between individuals and the natural world.

Our research makes the following contributions. First, we developed GreenCompass, a mobile game designed to help urban residents integrate nature engagement into their daily routines. By identifying pocket time, GreenCompass provides opportunities and guidance to connect with nature, making it easier for users to find moments of nature interaction in their busy urban lives. Second, we conducted a pilot study to gain preliminary insights on the user experience of GreenCompass which formed a basis for the future exploration of GreenCompass in promoting urban nature engagement. Finally, we hope this work can highlight the design opportunities for motivating human engagement with nature in urban areas and inspire designers who are interested in designing for human nature engagement to explore approaches that foster connections between people and the natural environment.

2 RELATED WORK

2.1 NatureHCI

In response to the increasing urbanization that leads to a lack of natural experiences and the ubiquitous influence of technology [40], the topic of nature has gained increasing attention in the field of human-computer interaction (HCI), leading to the emergence of several related subfields. Human-nature interaction (HNI) [40, 42, 49] focuses on the direct interaction between humans and natural elements, emphasizing the role of technology in mediating these interactions to strengthen connections with nature. Some theoretical perspectives take a posthumanist view, considering humans as integral parts of ecosystems. For example, collaborative survival [25, 26, 39] emphasizes the fundamental interdependence between human well-being and ecological health, and advocates resilient multispecies co-existence, particularly in ecologically stressed areas. The more-than-human perspective [3, 6, 48] redefines ecological systems by recognizing the agency and intrinsic value of non-human entities, and advocates a holistic approach that respects and integrates their roles alongside humans. Other theories emphasize interaction with particular elements of nature. For example, human-plant interaction [7] explores the interplay between humans and plants, investigating how design and technology can enhance this interaction. Human-animal interaction [18, 30] focuses on using technology and design to facilitate deeper human-animal interactions, improve understanding, improve animal welfare, and promote harmonious co-existence. In addition, the field of sustainable HCI [11, 15, 35] focuses on designing and using computing technologies to minimize environmental impact and promote pro-environmental behaviors. It intersects with topics such as energy use [22], resource consumption [16], and recycling [8], while some theories within this domain emphasize the incorporation of ecological principles into technology design and user interactions to foster harmonious relationships between people and nature.

Building on these theoretical foundations, researchers have developed frameworks to guide design practices. For example, Forest Playfulness [2] introduces a framework for designing playful technologies to enhance interactions with forests, promoting joy and sustainable engagement with nature. Similarly, Michael et al. [21] proposed a framework for designing interactive technology aimed

at improving natural recreation. In addition to exploring theoretical frameworks, efforts have also begun to investigate NatureHCI from a practical perspective. For example, GreenHat [36] is an interactive mobile learning application that enhances users' understanding of biodiversity and sustainability by providing real-time expert commentary on their surroundings, thereby increasing awareness of nature and deepening engagement with local ecosystems. This inspired us to explore ways of guiding users' attention to nature while also encouraging direct engagement with their surroundings. JeL [45], a virtual reality installation, enables users to synchronize their breathing with the growth of a coral-like structure, fostering a mindful experience which inspired us to explore the design of nature-related mindful activities to increase engagement with nature. Liu et al. [26] designed a glove that enables users to engage with soil and fungi, offering a tactile experience that strengthens their connection to the environment. This work inspired us to design interactions that help users explore natural spaces and foster a deep understanding of their surroundings. In addition, GoChirp [24] is an AI-powered wearable that uses haptic feedback from bird chirping detection to heighten users' awareness of urban wildlife and deepen their connection with nature, inspiring us to consider technology as a bridge to guide users towards active engagement with nature.

Despite existing efforts in NatureHCI, little work has been done to provide motivation to interact with nature in urban settings, where time constraints and busy routines often prevent regular access to nature [50]. Therefore, the purpose of our work is to bridge this gap by designing GreenCompass that inspires effortless and continuous nature engagement within urban contexts.

2.2 Outdoor Play in Nature

Outdoor play involves activities that integrate elements of the outdoor environment to promote creativity, social interaction, and physical activity [17]. In HCI, research has extensively explored two prominent approaches to enhancing nature engagement: pervasive games [29] and digitally enhanced natural playgrounds [20].

Pervasive games blend digital gameplay with physical environments, using technologies such as GPS and augmented reality (AR) to create "hybrid spaces" that extend beyond the screen [9]. These games encourage players to interact directly with their surroundings through movement and location-based activities [38]. For example, Geocaching [33] uses GPS technology to guide players to hidden containers, known as "geocaches," encouraging exploration and engagement with both urban and natural environments by motivating participants to discover new places and appreciate their surroundings. Similarly, Jurassic World: Alive [27] encourages players to locate and capture virtual dinosaurs at physical locations, motivating visits to outdoor spaces such as parks and trails. From these, we learned that location-based interactions can serve as effective prompts to encourage users to explore outdoor environments with a sense of curiosity and discovery. In addition, initiatives like using augmented reality in national parks to provide contextual information about flora, fauna, and geology have shown promise in enhancing users' educational and exploratory experiences in nature environments [14, 28]. These pervasive games encourage

playful exploration of real-world environments and can be adapted to natural settings to promote engagement with nature.

In comparison, digitally enhanced nature as playgrounds introduces technology directly into natural spaces [12]. These environments feature interactive installations with sensors, lights, and sound systems that respond to players' actions [5, 10, 20, 34, 37, 47]. For example, the "Animazia" play area at Wildlands Adventure Zoo [1] uses interactive projections and sensors to simulate natural phenomena, encouraging children to explore and engage with nature-themed environments in a playful and educational manner. Similarly, the Story-Informed Playground [20] uses narrative storytelling to encourage children to explore and engage with their natural surroundings. By embedding interactive and narrative-driven elements into natural landscapes, this approach aims to harmonize technology with the essence of nature, transforming outdoor spaces into dynamic environments for exploration and learning [4].

Pervasive games and digitally enhanced playgrounds illustrate two complementary approaches to enriching outdoor play with digital technology. GreenCompass extends this work by moving beyond designated play spaces to integrate playful nature engagement into everyday life, focusing on making nature interaction a sustainable daily practice rather than a destination-based activity.

3 GREENCOMPASS

3.1 System Design

In this work, we propose GreenCompass, a mobile social game that encourages players to engage with nature during their pocket time. Players can form groups with their friends to complete nature-related tasks in the game. Masashi et al.'s [42] framework of human-nature interaction highlights that meaningful engagement goes beyond simple exposure, requiring an alignment of opportunity, motivation, and capability to foster sustained interaction. Drawing on this framework, our system design addresses these three critical factors essential for overcoming barriers to nature connection.

Addressing the opportunity factor, we recognized that urban residents often struggle to visit green spaces due to perceived time constraints in their fast-paced lives [50]. GreenCompass tackles both temporal and spatial barriers to create accessible nature engagement opportunities. The system analyzes users' self-reported schedule information to identify and optimize available pocket time for nature engagement, while using GPS technology to recommend nearby nature spots. To ensure feasibility, GreenCompass calculates the required travel time from users' current locations to recommended spots and evaluates potential disruptions, ensuring the identified pocket time is sufficient for both travel and the proposed activity. This comprehensive planning approach helps users seamlessly integrate nature engagement into their daily routines while minimizing the planning effort required.

To address motivation, which is crucial for converting opportunities into actual engagement, GreenCompass incorporates several design features. First, we designed experiential-oriented tasks, such as collecting fallen leaves or touching trees, to create engaging and meaningful nature experiences. Second, we implemented social features that enable group members to view collective task progress, share and appreciate nature moments through a like-based interaction system, and participate in collaborative activities, leveraging

peer support and social motivation. These features work together to create an environment that continuously motivates users to build and maintain their connection with nature.

To accommodate users with varying capabilities, GreenCompass employs an adaptive task recommendation system. For users with lower physical fitness levels as self-reported, GreenCompass recommends accessible activities such as bird watching in parks, while those with higher fitness levels receive more challenging recommendations like mountain hiking. In addition, GreenCompass continuously learns from users' interaction history to understand their task preferences, enabling increasingly personalized recommendations that match individual capabilities to facilitate engaging outdoor experience.

Through this integrated design approach addressing opportunity, motivation, and capability, GreenCompass creates a comprehensive system that systematically breaks down barriers to nature engagement in urban settings, working to foster lasting connections between individuals and the natural world.

3.2 Implementation

GreenCompass was developed using the UniApp¹ framework, enabling efficient cross-platform development and ensuring seamless operation on both iOS and Android devices. The front-end was built using Vue², creating a responsive user interface that enables real-time interaction with users. For the back-end, we used a serverless architecture that stores user data in a cloud database. This setup facilitates efficient data processing and real-time data synchronization, ensuring that users experience smooth and uninterrupted service. To accurately match tasks with the user's current context, we used the AMap API³ for location services. This integration ensures that the tasks are relevant and contextually appropriate based on the user's current location. Task scheduling is managed by a rule-based constraint optimization algorithm. This approach allows for the dynamic task assignment, ensuring that each user's experience is tailored to their specific circumstances and preferences.

4 PILOT STUDY

To assess the usability of GreenCompass while gathering preliminary insights into its impact on nature engagement, we conducted a one-week pilot study with 3 participants (2 females and 1 male, no non-binary or self-described: age: $M = 22$; $SD = 1$) as a group and they were already familiar with each other. All participants were graduate students who were informed that the entire experiment would be recorded and provided their consent. We hypothesized that, after using GreenCompass, participants would (H1) show an increased sense of connection to nature and (H2) show an increase in self-reported frequency and duration of outdoor activities compared to baseline. Beyond these hypotheses, we also sought to understand participants' overall experiences with the system.

Before the study commenced, each participant completed the Nature Relatedness Scale (NR-6) [32], a six-item questionnaire rated on a five-point Likert scale, to assess their baseline connection to nature. Participants also reported their baseline time and frequency

¹<https://en.uniapp.dcloud.io/>

²<https://vuejs.org/>

³<https://lbs.amap.com/>

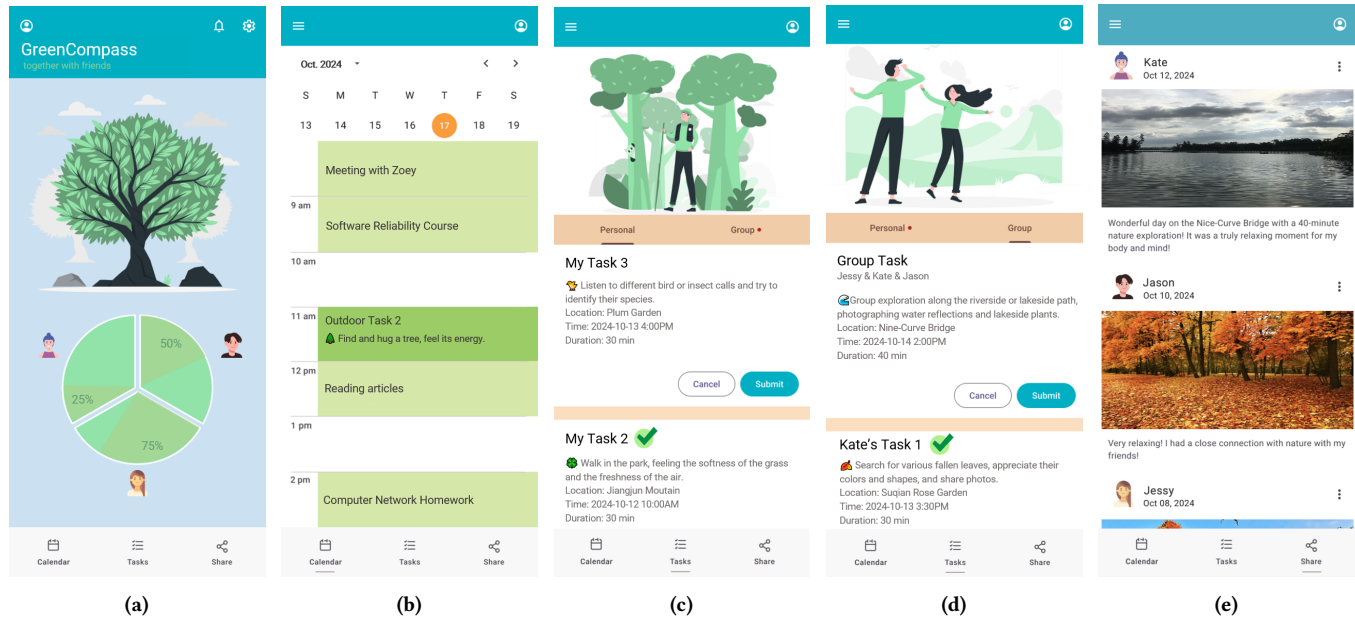


Figure 1: These five figures illustrate the interfaces of GreenCompass. (a) GreenCompass’s main page, with a shared progress bar showing players’ progress in completing nature-related tasks. (b) GreenCompass’s calendar page where players can schedule and receive tasks that use their free time. (c) GreenCompass’s personal task page, which displays individual outdoor tasks. (d) GreenCompass’s group task page, which displays the group task and the tasks completed by other members. (e) GreenCompass’s sharing page where players can share their nature-related experiences.

spent outdoors. The study then began with a short online tutorial designed to familiarize participants with the GreenCompass interface and its core functionalities, such as submitting tasks and sharing experiences. This step was crucial to ensure that all three participants had a basic understanding of how to navigate and use GreenCompass. Following the tutorial, participants engaged in a week-long period of unrestricted use of GreenCompass. During this phase, each participant was assigned three individual and one collective nature-related task from GreenCompass.

At the end of the study, participants completed the post-experiment NR-6 questionnaire and reported their time and frequency spent outdoors during the previous week. In addition, semi-structured interviews were conducted with each participant to explore their overall experience with GreenCompass. Each interview lasted 30 minutes and was designed to obtain changes in nature engagement, satisfaction with the system, and other relevant feedback from participants. Throughout each interview, notes were taken and audio recordings were made to ensure accuracy and to facilitate subsequent transcription.

4.1 Results

Overall, participants appreciated GreenCompass and said that it had brought them closer to nature. They found GreenCompass innovative and engaging, helping them overcome barriers to accessing the natural environment.

The results of the NR-6 questionnaire showed that participants’ sense of connection with nature increased over the week, with the average score rising from 2.28 ($SD = 0.29$) to 2.78 ($SD = 0.06$). This

suggests that GreenCompass played a positive role in increasing their connection to nature. Participants achieved an average task completion rate of 83.33% ($SD = 0.01$). In addition, their average outdoor time increased from 2.78 ($SD = 0.48$) hours to 3.55 ($SD = 0.26$) hours, while the frequency of outdoor activities increased from 3.33 ($SD = 0.33$) to 5.67 ($SD = 1.33$) times per week. These results show a positive trend in participants’ engagement with nature over the study period.

Participants reported that GreenCompass provided them with opportunities to explore nearby nature and they appreciated the local recommendations, which required minimal effort and time to reach and deepened their connection to the nature around them. For example, P2 said: “*The location suggestions were great. They showed me natural places nearby so I could avoid long journeys and get back to work or home easily.*” P1 added: “*The system seemed to understand when I had short breaks. It recommended tasks that fit seamlessly into those short intervals, eliminating my excuse of not having time to go outdoors.*”

Participants expressed that their motivation to engage with nature was significantly boosted by the social design of GreenCompass. Features like shared progress bars, collaborative tasks, and social sharing options made them feel part of a community. For instance, P1 said: “*The way the system connects us through shared goals and progress really increases my motivation to get out there. Knowing that I am part of a community working towards the same goals makes the whole experience more rewarding and less lonely.*” This sense of belonging provides participants support and encouragement from peers, making tasks less lonely and more engaging.

Participants appreciated the inclusivity of GreenCompass and described tasks that were well suited to their preferences in terms of style and physical demands. This customization reduces stress and allows users to feel physically active as they interact with nature. For example, P3 said: *“The tasks were adapted to my physical abilities and kept me accessible and never discouraging.”* P2 also said: *“The system offered activities that matched my interests, such as quiet walks in the park. This made it feel like it was made just for me and increased my desire to connect with nature.”* Participants appreciated this customized approach, as it made the nature experience enjoyable and accessible.

Participants enjoyed GreenCompass as it provided many tasks that involved outdoor activities they had never tried before, resulting in novel experiences of nature engagement. For example, P1 said: *“The tasks encouraged me to try activities I had never considered before, such as birdwatching or identifying native plants, transforming my usual outdoor time into exciting new adventures.”* Furthermore, all participants appreciated mental healing from continuous interaction with nature, reducing stress and improving emotional well-being. For instance, P2 said: *“Spending time in nature provided a sense of spiritual healing that I had not anticipated. It significantly reduced my stress levels and made me feel at peace.”*

Some limitations of GreenCompass were also reported by participants. P2 expressed a desire for a one-click sharing feature upon task completion that would allow users to easily share their accomplishments. In addition, P2 and P3 suggested the inclusion of a task refresh option that would provide the ability to update tasks as needed. Currently, GreenCompass lacks these features, which could further increase user engagement and satisfaction.

5 DISCUSSION AND FUTURE WORK

Our results demonstrated that GreenCompass increased participants' time spent outdoors and increased their sense of connection to natural environments, supporting our hypotheses (H1, H2) of increased nature-relatedness and outdoor engagement. Participants valued GreenCompass as providing an enjoyable and intuitive experience that made nature engagement more accessible and seamlessly integrated into their daily lives. Although some limitations were noted, the overall findings align with our expectations, suggesting that GreenCompass holds promise in supporting nature engagement in urban settings.

These results highlight GreenCompass's effectiveness in transforming everyday routines into opportunities for nature engagement. By identifying and utilizing users' pocket time, the system provided structured yet flexible ways to engage with nearby nature. Beyond individual engagement, social interaction played a key role in sustaining motivation and reinforcing participation. Prior research has raised concerns that social interaction might detract from nature immersion, potentially reducing the restorative benefits of outdoor experiences [31, 44]. However, our findings suggest that when social features are designed to align with nature engagement rather than compete with it, they can serve as a motivating force. In GreenCompass, elements such as shared goals and peer accountability appeared to enhance engagement by fostering a sense of collective encouragement. This suggests that social features that provide external motivation, such as peer expectations, can act as

engagement drivers, reinforcing participation while maintaining immersion. In addition, our findings show that allowing space for exploration within task-based engagement might further enhance nature interaction. Unlike traditional game tasks that guide players towards predefined goals, GreenCompass embedded exploration into the tasks, encouraging users to interact with nature in a more self-directed and immersive way. Participants in our study appreciated the variety of tasks and the opportunity to engage in novel outdoor activities, suggesting that the introduction of adaptable task recommendations might help to maintain engagement. Although balancing guidance with opportunities for discovery appears to be beneficial, further research is needed to investigate how different levels of task adaptability influence continued participation.

Building on participant feedback and these initial implications, we plan to refine GreenCompass to better align with user needs before a broader evaluation. Enhancing task adaptability and optimizing social features will be key areas of focus, ensuring that engagement remains both flexible and intuitive. These refinements will form the basis for future iterations, helping to strengthen the GreenCompass's ability to facilitate sustainable engagement with nature.

While our results indicate a relatively positive user experience with GreenCompass, the study remains limited in scope. The small-scale nature of our study means that further investigation is required to better understand the broader impact and long-term effects of these design choices. To gain a more comprehensive perspective, our next step of research will focus on conducting a formal user study. We aim to recruit at least eight groups, ensuring gender balance, for a four-week game experience. The study will use questionnaires before and during each week of the study, and will conduct semi-structured interviews at the end of the study. The questionnaires will include the NR-6 [32] to measure participants' changes in the connection of participants with nature, and the Outdoor Activity Scale to track changes in time and frequency of outdoor activity. Quantitative data will be analyzed using statistical methods, while interview data will be subjected to thematic analysis. Based on the findings, we will describe the user experience of GreenCompass and discuss the design implications for gamifying the engagement of urban nature. To further assess whether GreenCompass fosters sustained nature engagement rather than short-term behavioral changes, a delayed follow-up study will be considered as a future step. Our ultimate goal is to propose feasible design recommendations for future systems aimed at fostering the connection between urban residents and their natural environment.

6 CONCLUSION

In this work, we present GreenCompass, a mobile game designed to encourage players to engage with nature during their pocket time by forming groups with friends to complete nature-related tasks. GreenCompass inspires people to reconnect with nature by considering time constraints to provide opportunities for users, using gamification and social interaction elements to encourage participation, and adapting to people of different abilities to reduce barriers to interacting with nature in the city. Our pilot study revealed that this game effectively initiates players to connect with nature, helps them overcome barriers to accessing natural environments, and

provides players with the novel experience of engaging with nature. We hope that our work will inspire further design opportunities to promote human engagement with nature in urban areas, and encourage designers interested in designing for human engagement with nature to explore ways to foster connections between people and nature.

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